a

1 Tabla de integrales indefinidas

$$\int cf(x)\,dx = c\int f(x)\,dx \qquad \int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$$

$$\int k\,dx = kx + C$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1) \qquad \int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C \qquad \int b^x\,dx = \frac{b^x}{\ln b} + C$$

$$\int \operatorname{sen} x\,dx = -\cos x + C \qquad \int \cos x\,dx = \operatorname{sen} x + C$$

$$\int \sec^2 x\,dx = \tan x + C \qquad \int \csc^2 x\,dx = -\cot x + C$$

$$\int \sec x \tan x\,dx = \sec x + C \qquad \int \csc x \cot x\,dx = -\csc x + C$$

$$\int \frac{1}{x^2+1}\,dx = \tan^{-1}x + C \qquad \int \frac{1}{\sqrt{1-x^2}}\,dx = \operatorname{sen}^{-1}x + C$$

$$\int \operatorname{senh} x\,dx = \cosh x + C \qquad \int \cosh x\,dx = \operatorname{senh} x + C$$

5.4 EJERCICIOS

1–4 Verifique mediante derivación que cada una de las fórmulas siguientes es correcta.

1. $\displaystyle\int \frac{x}{\sqrt{x^2+1}}\,dx = \sqrt{x^2+1} + C$

2. $\displaystyle\int \cos^2 x\,dx = \tfrac{1}{2}x + \tfrac{1}{4}\operatorname{sen} 2x + C$

3. $\displaystyle\int \tan^2 x\,dx = \tan x - x + C$

4. $\displaystyle\int x\sqrt{a+bx}\,dx = \frac{2}{15b^2}(3bx - 2a)(a+bx)^{3/2} + C$

5–18 Obtenga las integrales indefinidas generales.

5. $\displaystyle\int (x^{1.3} + 7x^{2.5})\,dx$

6. $\displaystyle\int \sqrt[4]{x^5}\,dx$

7. $\int \left(5 + \frac{2}{3}x^2 + \frac{3}{4}x^3\right) dx$

8. $\int \left(u^6 - 2u^5 - u^3 + \frac{2}{7}\right) du$

9. $\int v(v^2 + 2)^2 \, dv$

10. $\int \sqrt{t}(t^2 + 3t + 2) \, dt$

11. $\int \dfrac{1 + \sqrt{x} + x}{x} \, dx$

12. $\int \left(x^2 + 1 + \dfrac{1}{x^2 + 1}\right) dx$

13. $\int (\csc^2 t - 2e^t) \, dt$

14. $\int \left(\dfrac{1 + r}{r}\right)^2 dr$

15. $\int (2 + \tan^2 \theta) \, d\theta$

16. $\int \sec t \, (\sec t + \tan t) \, dt$

17. $\int 2^t(1 + 5^t) \, dt$

18. $\int \dfrac{\operatorname{sen} 2x}{\operatorname{sen} x} \, dx$

19–20 Determine la integral indefinida general. Ilustre mediante una gráfica de varios miembros de la familia en la misma pantalla.

19. $\int \left(\cos x + \frac{1}{2}x\right) dx$

20. $\int (e^x - 2x^2) \, dx$